

Supervised Acute Lymphocytic Leukemia Detection and Classification Based - Empirical Mode Decomposition

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Abstract—Acute lymphocytic leukemia (ALL) is a potentially fatal disease that affects both children and adults. ALL spreads quickly in the human body and kills within weeks. Hematologists examine the blood and bone marrow to see if ALL is present. Manual blood testing methods, which have been used for a long time, might be slow. This study provided an advanced computerized mechanism for classifying leukemia into two types: normal cell leukemia and blast cell leukemia. To begin, advanced image preprocessing algorithms were used and applied to acquire data augmentation, enhancement, and transformation. Furthermore, the K-means clustering technique was used to accurately segment the appropriate nuclei from the background. Moreover, most salient features have been extracted via an empirical mode decomposition (EMD) based Hilbert-Huang Transform (HHT). The extracted features have been fed toward supervised classifiers in order to classify the nuclei. A neural networks (NNs) classifier with the Bayesian regularization (BR) method has been implemented. Moreover, support vector machine (SVM), KNN, Random forest, Naïve Bayes, Logistic regression, and Decision trees classifiers have been utilized, tested, and compared with NNs. Experimental results show that NNs outperformed a classification accuracy of 98.7%, sensitivity of 99.3 %, and specificity of 98.1%.

Keywords— ALL, k-means, Empirical mode decomposition, Neural network, Classification;

I. INTRODUCTION

Cancer is the second largest cause of mortality worldwide, according to a World Health Organization report (WHO). A rise of immature and non-functioning leukocytes of diverse sorts characterizes leukemia, a type of blood cancer. The recovery of affected youngsters is aided by early identification of this disease [1]. ALL and AML are two forms of acute leukemia blood illnesses that must be diagnosed as soon as feasible. ALL is a kind of lymphoma that mostly affects the blood and bone marrow and affects a group of lymphocytes. ALL in children must be treated early because it spreads quickly to vital organs and bloodstreams, causing major health impacts [2]. Blast cells come in a wide range of forms. It's small, with homogenous

chromatin, microscopic nucleoli, and scanty basophilic cytoplasm. It's also conceivable to have large nucleoli with varied chromatin. Large cytoplasmic vacuoles are found in some blast cells [3]. The leukemia laboratory test is difficult, time-consuming, and recurrent. Treatment for this illness varies, and counting blast cells can be misleading. Machine learning (ML) algorithms have been found to be highly efficient at separating blast cells from healthy cells, and they are acquiring favor because they are faster than traditional detection [4].

II. LITERATURE REVIEW

With regard to medical knowledge and CD-marker comparisons, Jakkirich Laosai et al. [5] have suggested a method for classifying acute leukemia. Blood cells are classified into three categories: ALL, AML, and healthy, with sub-types for each classed cell group based on unique features information about medical experience, knowledge, and abilities understanding. Sonali Mishra et al. [6] have used the Y component of the CMYK image and a triangle thresholding technique to pre-process the input photographs. Texture properties were extracted using the discrete orthonormal S-transform (DOST), and dimensionality was reduced using linear discriminant analysis. The ADBRF classifier, which uses the random forest as a classifier, has the following properties. Christian Marzahl et al. [7] have suggested a method for "Classification of Normal versus Malignant Cells in B-ALL White Blood Cancer Microscopic Images" in order to avoid overfitting. This method employed innovative techniques to segregate and apply the theory at different learning rates. To improve their outcomes, they added a cursory attention method based on a region suggestion subnetwork for classifying ignorant abnormal and normal cells. Salman Shah et al. [8] have employed their own deep learning system. They employed a discrete Fourier transform in conjunction with an RNN to control the spectral features of the cells in their model, which is made up of a group of convolutional and recurrent neural networks. Pradeep Kumar et al. [9] have developed a robotic system for classifying and

identifying objects (ALL). For segmentation, the K-means approach was utilized. For feature extraction, the Gray-level co-occurrence matrix was used. Fully linked layers and/or dropout layers are used in the ResNet50 architecture. Ayush Pradhan et al. [10] have built three convolution learning-based models during the feature extraction stage. Logistic regression, SVM, and random forest classifiers were used to classify this model. Laura Bold et al. [11] have developed a deep learning-based method to monitor the diagnosis and management of leukemia using blood cell images utilizing the ALnet platform.

III. PROPOSED METHODOLOGY

A strategy for classifying acute lymphocytic leukemia was proposed in this research. Pre-processing, segmentation, feature extraction, and classification are all required for ALL classification. The proposed technique is depicted in Fig. 1.

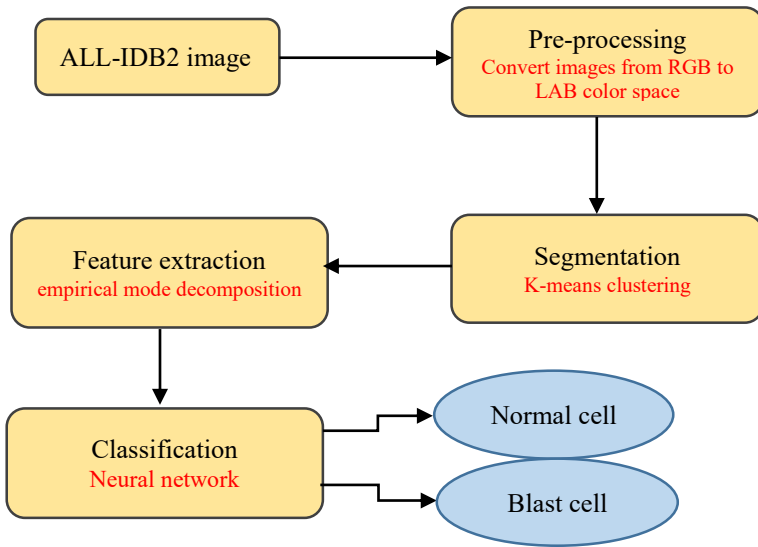


Fig. 1. Algorithm proposed for ALL classification

IV. DATA ACQUISITION

The initial step in any system is image acquisition. The main purpose is to obtain more visible and larger images. The ALL-IDB2 dataset was used in this work, which has a resolution of $257 \times 257 \times 3$ pixels [12]. As seen in Fig. 2, it generates 1100 pictures, 546 of which are normal cells and 554 of which are blasts. Then, to keep expanding the number of images, some data augmentation was applied.

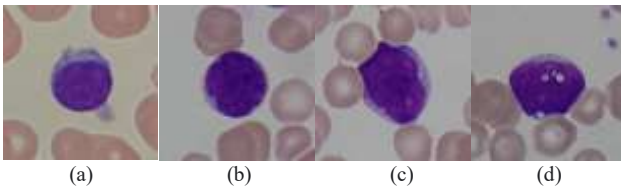


Fig. 2. (a) healthy cell, (b) L1 blast cell, (c) L2 blast cell, (d) L3 blast cell.

A. Data Augmentation

This phase was employed to increase the number of images in order to acquire better results. Rotation, brightness, zooming, shearing, and translation are all examples of data augmentation. In this investigation, several data augmentation strategies were applied.

- Rotation: An angle ranging from 0 to 360 degrees is used to rotate the image clockwise from the center. Augmentation is done at 90 and 180 degrees.
- Translation: An image is shifted in one of two directions: X or Y, or both. In this research, the image was translated by moving it 15 pixels in the X-direction and 25 pixels in the Y-direction, then 25.3 pixels in the X-direction and -10.1 pixels in the Y-direction.

Fig. 3. Show data augmentation results.

V. PRE-PROCESSING

A. Convert images from RGB to LAB color space

The majority of images have three color components. Because digital microscope images are commonly in RGB color space, segmentation is difficult. In reality, a number of factors such as image quality, brightness, and an older stain can cause the hue and intensity of blood cells and the image background to change dramatically. An adaptive technique is frequently utilized for making cell segmentation more effective in the face of these instability, reduce storage needs, and improve processing time [13]. Color space was converted from RGB to LAB. This color space has a brightness layer L that defines the color's lightness, as well as color space layers a and b that show the color's position between red, magenta, green, yellow, and blue, as illustrated in Fig. 4. The a and b layers were also employed for nucleus segmentation because they contain all of the color information.

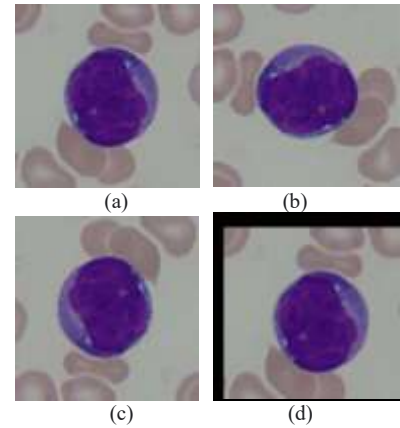


Fig. 3. (a) Original image, (b) image rotation 90 degrees, (c) image rotation 180 degrees, (d) image translation

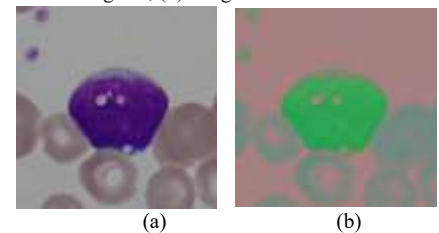


Fig. 4. (a) RGB image, (b) LAB image

VI. SEGMENTATION

Based on the features of participation, the segmentation process separates an image into separate regions. To distinguish between normal and blast cells in this study, the nuclei must be segmented from the image. This step includes the implementation of the K-means clustering technique.

A. K-means clustering algorithm

For segmenting the nucleus of lymphocytic cells, the K-means clustering approach is recommended. The unsupervised learning algorithm K-Means clustering appears to have no labelled data, unlike supervised learning. K-Means groups objects with similar features and distinguishes them from those in other clusters. The letter 'K' stands for cluster count. Nuclei are gathered in this work by assembling images into three color clusters. Using k-means clustering, the objective function is minimized. It can be rewritten as a method's objective function.

$$J = \sum_{j=1}^k \sum_{i=1}^n \|x_i^{(j)} - c_j\| \quad (1)$$

Where J denotes the spacing between the pixel (x) and the cluster center (c), k means the number of clusters, and n represents the number of data. The procedures used to segment nuclei from images using k-means are represented in Algorithm (1). Fig. 5, Fig. 6, and Fig. 7 show the processes for applying the suggested to abnormal images in several types of blast cells, as well as how well the blast cells were segmented. Fig. 8 and Fig. 9 show the steps in normal images. In this step, leukocyte nuclei were isolated using color-based clustering segmentation. Each pixel of an item has two values in reality: a and b. Because the main goal is to divide the image into three sections, each pixel is divided into nucleus, other cells, and background stain based on these two values. Each pixel in the Lab colour space has been classified into one of the three clusters using Equation (1) to measure the spacing between the pixel and each colour indication, resulting in each pixel of the image sequence being labelled to a particular colour based on the minimum distance from each indicator. This study focused on the cluster that contains a nucleus. According to the results of the experiment, the nucleus cluster was the one with the least quantity of red color.

Algorithm 1: segmentation of nuclei from images

Step1: Read the image from the dataset
Step2: Transform an image from RGB to LAB color space.
*Step3: Classify the colors in a*b* Space using K-Means.*
To prevent local minima, repeat the clustering three times.
Step4: Through using K-means outcomes, classify each pixel in the image.
 for $k = 1: n_colors$
 color = he;
 color (rgb_label ~ k) = 0;
 segmented_images{k} = color;
 end
Step5: Create segmentation results based on color morphology.
Step6: Create a separate image for the nuclei.

VII. FEATURE EXTRACTION

Feature extraction is the process of converting an image into data such that a model can classify it into two groups. Using the EMD approach, the proposed scheme was able to extract nine features from images.

A. Empirical Mode Decomposition

The oxidation of non-linear and non-stationary signals that used a responsive process based as EMD obtains a set of variable frequency components known as intrinsic mode functions (IMFs) depending on the Hilbert-Huang Transform [13]. The core principle behind the procedure is to accurately detect these implicit oscillatory formats in the data based on their attribute time scales, and then break the data down correctly. Sifting [14] will be used to remove most of the riding waves, or oscillations with no zero crossing among extrema. As a result, the EMD technique considers signal oscillations at a very local level and separates data into non-overlapping time scale modules. To depict the signal, dissociate a data set $x(x,y)$ into IMFs $x_n(x,y)$ and a residual $r(x,y)$.

$$x(x,y) = \sum_n x_n(x,y) + r(x,y) \quad (2)$$

Provided a signal $x(x,y)$, the useful EMD algorithm is described in Algorithm (2).

Algorithm 2: features extraction using EMD

Step1: recognize all of the extremes of $x(x,y)$
Step2: alternate among minima (or maxima), actually results in an envelope $e_{min}(x,y)$ (resp. $e_{max}(x,y)$)
Step3: calculate the mean $m(x,y) = (e_{min}(x,y) + e_{max}(x,y))/2$
Step4: extract the detail $d(x,y) = x(x,y) - m(x,y)$
Step5: keep iterating on the residual $m(x,y)$

In exercise, the previous framework must be improved through a sifting process that implies iterating steps 1 to 4 on the detail signal $d(x,y)$ until the latter can be called zero-mean according to some stopping criterion [15].

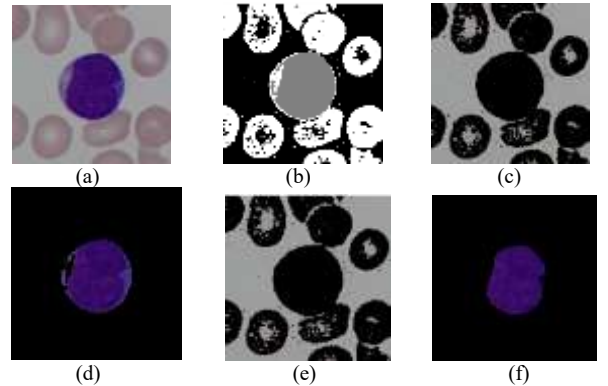


Fig. 5. (a) Original L1 blast cell, (b) Cluster index, (c) Objects in cluster1, (d) Objects in cluster2, (e) Objects in cluster3, (f) Segmented image

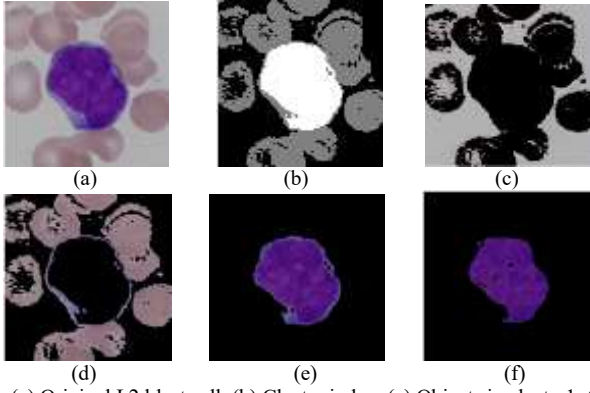


Fig. 6. (a) Original L2 blast cell, (b) Cluster index, (c) Objects in cluster1, (d) Objects in cluster2, (e) Objects in cluster3, (f) Segmented image

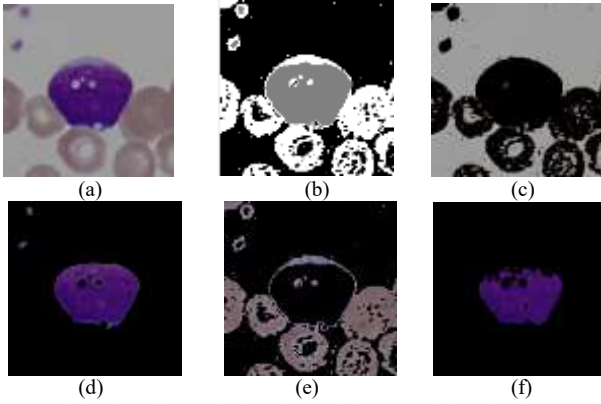


Fig. 7. (a) Original L3 blast cell, (b) Cluster index, (c) Objects in cluster1, (d) Objects in cluster2, (e) Objects in cluster3, (f) Segmented image

VIII. CLASSIFICATION PROCESS

The work's final step is classification. The major goal of this step is to discriminate between normal and blast cells after obtaining vector features for the diagnostic technique. A neural network classifier was used to classify normal and blast cells in this study.

A. Neural network classifier

In reality, the traditional classification method can be quantified by creating a stationary class-conditional probability distribution with a collection of fundamental functions. Classification is the most common use of neural networks. A well-known neural network architecture for classification is the multilayer perceptron. Because the network classifier contains nine inputs, it has been tweaked to provide the best results. The number of hidden neurons was limited to 15 neurons and one output in order to achieve the optimum results. The data is divided into three categories: 70% for training, 15% for testing, and 15% for validation. Using the BR function as the training function yielded the best results. The neural network's output appears to be either 0 or 1. A normal cell is represented by nil, while a blast cell is represented by one.

B. Bayesian Regularization(BR)

BR, a linear merger of Bayesian and NN methods for directly measuring optimum regularization parameters, is one of the regularization strategies in NN. One of the regularization strategies utilized in Bayesian regularized NN (BRNN) concepts is enforcing

multiple prior distributions on model parameters [16]. In order to achieve better generalization and softer mapping, BRNN provides an extra term E_w to the objective function of early stopping. A gradient-based optimizer, which works similarly to classical back-propagation, is then used to constrain the function. This method is equivalent to a penalized log-likelihood method. Modifying a frequently used objective function, including the mean sum of squared network errors (MSE or E_d), is part of the Bayesian Regularization method [17].

$$F = E_d = \frac{1}{N} \sum_{i=1}^N (e_i)^2 \quad (3)$$

Where N is the number of data points in the sample and $(e_i)^2$ is the square of the errors. The modification's focus is to enhance the model's generalization abilities. The objective function in Equation (3) is expanded by including a term E_w , which is the sum of squares of the network weights

$$F = \beta E_d + \alpha E_w \quad (4)$$

Where, E_d is a criterion for early stopping to avoid over-fitting, d : is the training set with input-target pairs, α and β are Bayesian parameters that need to be tuned. The factors are associated to the unknown variances associated with the weights and biases of the network. αE_w : The weight decay and α is coefficient of the weight decay, prefers small values of w and reduces the tendency of a model to over fit d .

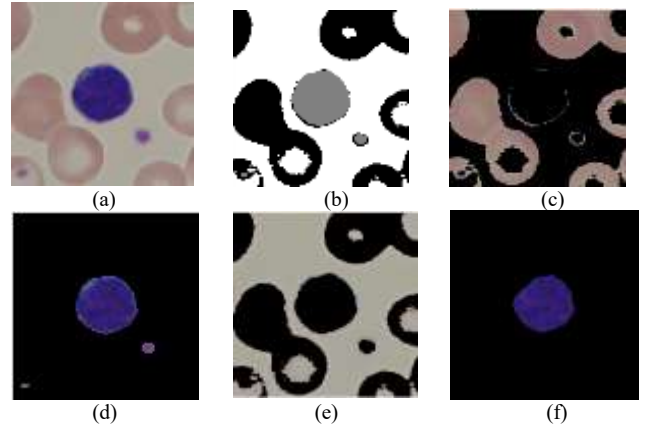


Fig. 8. (a) Original normal cell, (b) Cluster index, (c) Objects in cluster1, (d) Objects in cluster2, (e) Objects in cluster3, (f) Segmented image

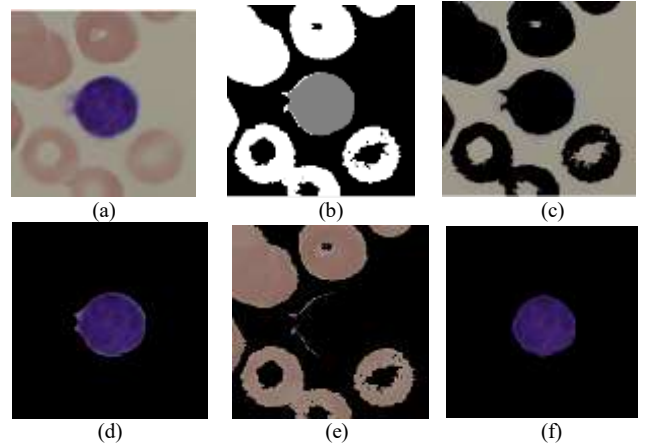


Fig. 9. (a) Original normal cell, (b) Cluster index, (c) Objects in cluster1, (d) Objects in cluster2, (e) Objects in cluster3, (f) Segmented image

IX. EXPERIMENTAL RESULTS

Traditional biopsy procedures are less successful than a computer-based early Acute lymphocytic leukemia diagnostic system. For the classification of ALL, the methodology employs machine learning and digital image processing so, both the cost and the time required for detection are reduce. To remarkably successful in both decision-making and pattern recognition applications, neural network classifier was used. The results of the tests were analyzed using MATLAB R2019a and a corei5 processor (8GB RAM). To train the model and classify the images as normal or blast, 554 blast images and 546 normal images from the ALL-IDB2 data set were used. The EMD method was used to extract the most important features from the images and feed them to a 15-layer neural network classifier. The data is divided into three categories: 70% for training, 15% for validation, and 15% for testing. The neural network training state is illustrated in Fig. 10. Fig. 11, shows the neural network training error histogram. When feature vectors calculated for the segmented region are fed into a neural network classifier, the BR algorithm produces the best results, with an accuracy of 98.7 %, sensitivity of 99.3 %, and specificity of 98.1 %. Fig. 12, shows the confusion matrix of the proposed classifier and other classifiers. Specificity (S), Accuracy (Acc), and Sensitivity (S) were calculated from these values. As shown in Fig. 12, the proposed classifier outperformed all other classifiers in the confusion matrix. Fig. 13, depicts the training ROC curve. Fig. 14, depicts the plot with the best training performance. The test and train curves fluctuate in a comparable manner, as shown in this figure, indicating that the network has been adequately modelled and trained.

A. Performance indicies

- Accuracy

The degree to which the classifier's detection is correct. To calculate accuracy, the following equation has been utilized.

$$Acc = \frac{TP + TN}{TP + FP + FN + TN} \quad (5)$$

Where, TP denotes true positive, TN denotes true negative, FN denotes false negative, and FP denotes false positive.

- Sensitivity

It evaluates the proportion of positive samples correctly identified by the proposed classifier.

$$S = \frac{TP}{TP + FN} \quad (6)$$

- Specificity

It computes the proportion of negative samples correctly identified by the proposed classifier.

$$S = \frac{TN}{TN + FP} \quad (7)$$

Table I displays the results of various classifiers, with the proposed classifier outperforming other classifiers in terms of accuracy, sensitivity, and specificity. When the results of various studies are compared, the proposed method based on empirical mode

decomposition outperforms other recent research studies of interest, as shown in Table II.

Table I. Comparative between classifiers

Classifiers	Accuracy	Sensitivity	Specificity
Logistic Regression	56.4%	58.6%	54.1%
Naïve Bayes	57.1%	20%	93.5%
Decision tree	62.9%	61.5%	64.2%
SVM	69.3%	73.4%	65.1%
KNN	72.5%	68.3%	76.7%
Random forest	76.5%	67.5%	71.4%
Proposed	98.7%	99.3%	98.1%

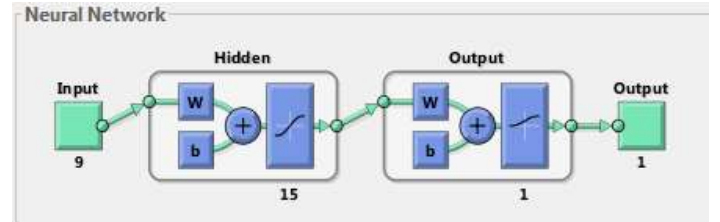


Fig. 10. The neural network training schematic diagram

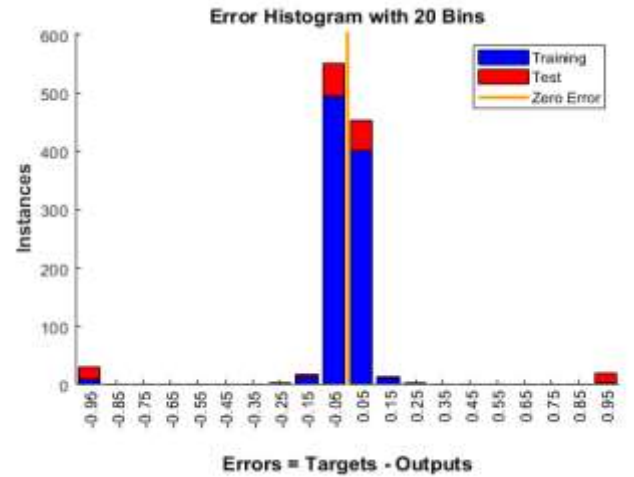


Fig. 11. Neural network training error histogram



(a)



(b)



(c)



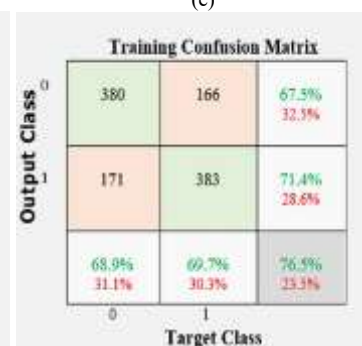
(d)



(e)



(f)



(g)

Fig.12 (a) confusion matrix of NN classifier, (b) confusion matrix of Logistic Regression classifier, (c) confusion matrix of Naïve Bayes, (d) confusion matrix of Decision tree classifier, (e) confusion matrix of SVM classifier, (f) confusion matrix of KNN classifier, (g) confusion matrix of Random forest classifier

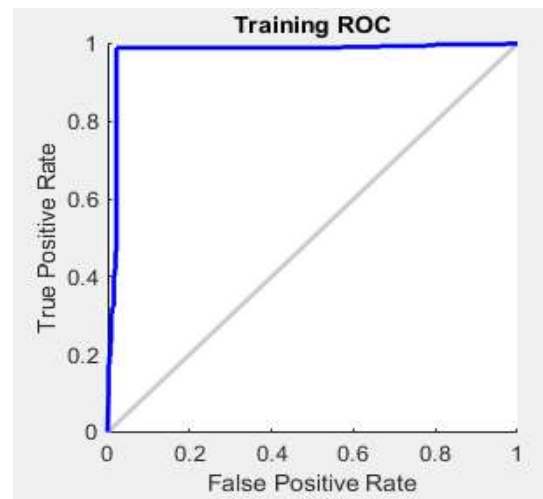


Fig. 13. ROC training curve

The performance plot is an extremely useful tool for monitoring how the network's mean square error decreases over time. The best validation performance at epoch 9 was 0.31318. This is depicted in Fig. 14, where the circle indicates a high training performance.

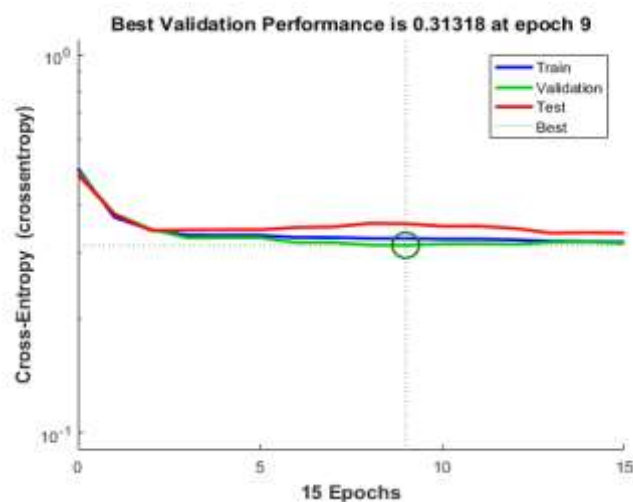


Fig. 14. The Best performance plot of the training of NN classifier, the circle represents the best training performance

Table II. Classification performance for different studies

Author, Year	Dataset	Pre-processing	Segmentation	Feature extraction	Classifier	Acc
(Sreeja Kochuvila et.al 2021)	ALL-IDB	Convert images from RGB to LAB	K-means clustering	area, perimeter, eccentricity, circularity, and sodality	SVM	95%
(Pradeep kumar et. al 2020)	ALL-IDB1	CLAHE method	K-means clustering	Gray-level co-occurrence matrix (GLCM) and gray-level run-length matrix (GLRLM)	SVM	96%
(Deepika Kumar et. al 2020)	SN-AM	Normalization, multicollinearity, scaling and data augmentation	Detected Edges	SelectkBest algorithm	Convolutional neural networks (CNN)	97.2%
(Ahmed T. Sahlol et. al 2018)	ALL-IDB2	Convert images from RGB to CMYK	Zack method	Social Spider Optimization algorithm (SSOA)	KNN	95.2%
(Sachin Kumar et. al 2018)	Dr RML Awadh hospital, Lucknow	Wiener filter, histogram equalization, and gray-scale transformation	k-means clustering	Color, Geometric, Texture and, Statistical	Nearest Neighbor (KNN) and Naïve Bayes	92.8%
(Amjad Rehman et. al 2018)	ALL-IDB	Convert images from RGB to HSV	Simple threshold	LBP method	Convolutional neural networks (CNN)	97.7%
Proposed method	ALL-IDB2	Data augmentation, and Convert images from RGB to LAB	K-means clustering	Empirical mode decomposition	Neural network with BR algorithm	98.7%

X. CONCLUSION

Acute lymphocytic leukemia classification methodology has been proposed in order to accurately discriminate between their two main classes; normal and blast. List of image processing algorithms have been utilized and implemented as a preprocessing stage. Moreover, K-means clustering approach has been employed in order to accurately segment foreground nuclei from its background. The extraction of salient features from segmented images have been done using empirical mode decomposition taking into consideration the sifting algorithm. The proposed method is intended to aid laboratory

professionals and pathologists, and it has a significant clinical impact on Leukemia patients.

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